MICHAEL P. RAGSDALE

Senior Full Stack Engineer | Web Platform Architect Headless CMS Architecture • Core Web Vitals • Accessibility Engineering (WCAG) Norfolk, VA | U.S. Citizen | hireme@michaelpragsdale.com

Professional Summary

Senior Full Stack Engineer specializing in architecting highly scalable, public-facing web platforms, headless content management systems, and accessibility-first user experiences. Creator of the Stardust Engine CMS, a zero-database JSON architecture powering a multi-domain ecosystem through Cloudflare edge routing. Passionate about bridging the gap between marketing, product, and engineering by building lightning-fast, semantic, and AEO-optimized web experiences that drive discovery and conversion.

Technical Skills

- **Web Platform Architecture:** Headless CMS Design, Server-Rendered Architectures, JSON Routing, REST APIs, SPA Development, Answer Engine Optimization (AEO)
- **Frontend Technologies:** Vanilla JavaScript (ES6+), HTML5, CSS3/SCSS, Responsive Component Design, Web Performance Monitoring (Core Web Vitals)
- **Accessibility Engineering:** WCAG 2.1 AA, Section 508, Semantic HTML, NVDA/JAWS, Focus Management
- **Infrastructure & Tooling:** Cloudflare (DNS/CDN), Nginx, DigitalOcean, Bash, Python, Git, CI/CD Pipelines

Relevant Technical Experience

Principal Systems Architect & Founder

RaggieSoft Media | Norfolk, VA | raggiesoftmedia.com | Est. 2008 – Present Founded and operate RaggieSoft Media, an independent software development and digital media publishing ecosystem supporting web applications, cloud infrastructure, narrative publishing, music distribution, and open-source platform development.

- **Platform Architecture & Headless CMS:** Architected and open-sourced **Stardust Engine CMS**, an MIT-licensed, zero-database PHP MVC backend powering a 9-domain ecosystem (including `raggiesoft.com` and `stardustenginecms.com`), decoupling content models from presentation.
- **Multi-Domain & Cloud Edge:** Orchestrate a production fleet of 9 distinct domains via **Cloudflare DNS/CDN** edge routing, 301 redirection rules, and automated deployment pipelines syncing assets to **DigitalOcean Spaces** (S3-compatible object storage) and Nginx reverse proxies.
- **Web Performance & AEO:** Engineered JSON-driven content models and a centralized Vanilla JS rendering engine optimizing Core Web Vitals (sub-second LCP), technical SEO, and Answer Engine Optimization (AEO) for downstream AI discovery.
- **Automated Media Engineering:** Engineered automated server-side pipelines utilizing **FFmpeg** and Bash to transcode and normalize audio assets across multiple formats (FLAC, WAV, MP3, OGG), paired with custom **PHP GD** engines that dynamically composite Open Graph social cards across routes.
- **E-Commerce & Digital Fulfillment:** Integrated **Fourthwall** as an international Merchant of Record (MoR) with custom DNS architecture, powering automated, frictionless global fulfillment of physical merchandise and digital assets without third-party vendor lock-in.
- **Accessibility by Design:** Enforce strict **WCAG 2.1 AA** and **Section 508** compliance across all UI components, custom modals, media players, and narrative readers, ensuring zero accessibility debt out of the box.

Accessible Document Engineer (Clerk-Cashier)

City of Virginia Beach Parks & Recreation | Virginia Beach, VA | May 2023 – Present

- **Data Engineering & ETL:** Architect manual ETL workflows to extract and normalize complex operational data from legacy enterprise platforms (ActiveNet) into highly structured, digital formats.
- **Compliance Enforcement:** Govern output data to guarantee strict Section 508 compliance across generated operational documents, ensuring institutional adherence to federal accessibility mandates.

Web Accessibility Specialist (Computer Operations)

Old Dominion University (ODU Online) | Norfolk, VA | Nov 2019 – Jul 2022

- **Enterprise Web Quality:** Served as the primary technical gatekeeper for web content quality and accessibility during a massive, university-wide digital infrastructure migration to the Canvas LMS.
- **WCAG Remediation:** Executed rigorous manual testing and remediation of complex HTML/CSS codebases to ensure strict structural integrity, semantic correctness, and compatibility with screen readers.
- **Architectural Standardization:** Engineered universal, inherently accessible architectural templates deployed across the university ecosystem, significantly lowering the cognitive load for content creators while maintaining strict baseline compliance.

Education & Certification

- **Tidewater Community College | Virginia Beach, VA** Associate of Science, Information Technology (Software Engineering) | Expected Dec 2027
- **Old Dominion University | Norfolk, VA** Bachelor of Science, Customized Studies in Leadership | Prior Coursework
- **Wilson Workforce and Rehabilitation Center | Fishersville, VA** Certificate, Business & Information Technology | 2011